

Milestone 3 – User Feedback Report

1. Feedback Questions and Evaluated Aspects

To evaluate the usability and effectiveness of our dashboard, we collected feedback from two peers enrolled in the DATA 551 course. The feedback process was informed by the System Usability Scale (SUS) framework and adapted to the goals and target audience of our dashboard.

The feedback questions focused on the following aspects:

User Background

- User role (e.g., student or researcher)
- Familiarity with interactive data visualization dashboards

Ease of Navigation

- Whether the dashboard layout was intuitive
- Ease of navigating between filters and visualizations

Interactivity and Usability

- Clarity and usefulness of interactive filters (metric selection, year slider, region and country filters)
- Intuitiveness of interactive elements such as dropdowns and hover tooltips

Visualization Clarity

- Effectiveness of the charts in communicating trends and comparisons
- Clarity of titles, axis labels, and legends

Performance and Accessibility

- Responsiveness of the dashboard during interaction
- Any accessibility or readability concerns

Expectations and Suggestions

- Features users expected but did not find
- Suggestions for improving usability or functionality

These questions allowed us to assess overall user experience, usability, and visual clarity of the dashboard.

2. Summary of User Feedback

Overall, users reported a positive experience using the dashboard.

Positive Feedback

- Users found the dashboard layout clean, organized, and easy to navigate.
- The separation between the filter panel, visualization area, and legend/help panel was considered clear and effective.
- The combination of bar chart, scatter plot, and line chart was helpful in understanding both cross-sectional comparisons and temporal trends.
- Interactive filters were described as intuitive and responsive.

Identified Issues and Limitations

- One user noted that the number of available filters could feel overwhelming at first glance.
- Users mentioned that when certain filter combinations resulted in limited data, the fallback behavior was not always obvious.
- Color usage across plots could be improved for better consistency and accessibility.
- One user expected brief explanatory text to help interpret the charts, particularly for first-time users.

3. Planned Improvements Based on User Feedback

Based on the collected feedback, we plan to implement the following improvements in the final milestone:

1. Improve user guidance

We will add short explanatory text or tooltips to help first-time users understand how to interact with the dashboard and interpret the visualizations.

2. Refine filter presentation

To reduce cognitive load, advanced or less frequently used filters will be visually grouped or collapsed by default.

3. Enhance visual consistency and accessibility

We will refine color schemes across all plots to improve consistency and consider accessibility, including color-blind friendly palettes.

4. Clarify fallback behavior

When fallback logic is triggered due to sparse data selections, we plan to provide clearer visual or textual cues to indicate which data is being displayed.

Some suggested features, such as exporting data or saving customized views, are beyond the scope of this course project and will not be implemented due to time constraints.